Chapter 7
Artificial Intelligence-Based Computer Vision on Medical Images

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ABSTRACT

The trajectory in medical imaging is characterized by a growing volume of data that is increasingly considered an inherent demand across various medical disciplines. With advancements in computer technology has become swifter, resulting in a surge of data for physicians to analyze. Artificial Intelligence (AI) falls within the realm of computer science. Its primary goal is to create machines capable of cognitive functions such as perception, learning, and problem-solvingthat surpass human capabilities. AI encompasses the spectrum of computer technology that displays characteristics reminiscent of human intelligence. Computer vision applied to medical images is an influential domain that utilizes various methods, including image processing and deep learning, to analyze, interpret, and extract valuable insights from medical imagery.

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1. INTRODUCTION

Computer vision has advanced significantly in the last decade as a consequence of AI applications that use machine learning (ML) methods. As a result of these advancements, the medical industry has been greatly impacted. AI-powered systems that improve medical image processing, streamline clinical procedures, and assist in decision-making have been developed (Bhatt C. et al. 2021). Promising results have been shown by many investigations and publications that use AI and ML in a variety of medical applications (Singh R 2020; Wang M et al. 2020; Wang C et al. 2019; Litjens G et al. 2017; Wang T et al. 2021; Thompson RF et al. 2018; Hosny A et al. 2018). With recent advances in AI, tasks including outcome prediction, image segmentation, and disease diagnosis have experienced revolutionary shifts. Recently, machine learning tools have matured to the point where they meet clinical needs. As a result, research teams, organizations, and medical professionals are working together to develop AI solutions for clinical applications. The state of affairs now puts the application of AI in clinical contexts closer than ever. Thus, it is now crucial for those working in the medical industry to comprehend the foundations of this technology.

Large amounts of medical picture data could be handled by ML developments. These techniques are helpful, but they also have some drawbacks (Lee JG et al., 2017). The necessity of manual feature extraction and selection, which is essential for determining important factors that correspond with outcomes, is one of the main issues. In addition, for artificial intelligence algorithms to be used in routine medical settings, they need to be dependable and fast in delivering outcomes within reasonable computation times. Deep learning (DL), the most current generation of machine learning, has a number of improvements over earlier forms of ML (Shi L et al. 2017). Deep learning (DL) uses numerous layers of neural networks, increasing neural intricacy and significantly increasing processing power.

1.1 Machine Learning

Machine Learning is regarded as a subfield of Artificial Intelligence as defined by Sedik A et al. (2020). Machine learning employs some set of logics that can be used by the computer to respond to future queries by learning a set of data called training set. The huge volume of data generated by various scientific domains has made AI more useful tool in assessing OMIC research (Min Set et al. 2017). The fundamental principle of machine learning involves generating precise forecasts for novel, unseen data subsequent to training on a finite dataset. In machine learning, simplifying input data to highlight crucial patterns is essential for optimal algorithm performance. The accuracy of these features' identification and extraction significantly

influences the learning algorithms' effectiveness. Feature engineering demands substantial time and expertise in domain-specific knowledge to customise data representations, making it a laborious and intricate process. On the contrary, deep neural networks (DNNs) (Kumar I. et al. 2022) independently construct complex data representations tailored to the task, eliminating the requirement for designing new feature extractors for every problem. However, a significant limitation is their reliance on large volumes of input data to yield optimal performance. Creating a model independently through a programme's automated process representation from a dataset isn't novel; Perceptron stands as one of the initial methods that envisioned conceptualising data directly from the environment.

1.2 Various Learning approaches

Different learning methods are present in the domains of machine learning (ML) and deep learning (DL), as highlighted in the works of Meyer et al. (2018). These methods are categorized into supervised, unsupervised, reinforcement, and transfer learning approaches. In supervised learning, the algorithms study the relationship between the input data and the corresponding output called class label. The knowledge gained from the relationship between the input and class labels is applied to determine the output/class label of an unknown data. These approaches consistently produce reliable results than other approaches. In contrast to supervise learning approach, the unsupervised model determines the innate structure of the given dataset rather than depending on the supplied output. These approaches employ response variable information for predicting diagnoses. Methods such as regression are used for continuous data points, while classification approaches are used for discrete groups. However, probability density estimation is typically utilized in unsupervised learning tasks like manifold learning, clustering, dimensionality reduction, and the identification of outlier information. While less prevalent than supervised learning, unsupervised techniques such as domain adaptation (Chen C et al. 2020; Perone CS et al. 2019), the generation of data (Liu Y et al. 2019; Liu Y et al. 2020), and segmentation of generated medical images (Aganj I et al. 2018) were extremely promising. The foundation of reinforcement learning is inquiry, and feedback is utilized to adjust the method of learning. In certain scenarios, including numerous medical applications, transfer learning proves helpful when there is insufficient input data to perform training. Using less labelled input to increase the network weights of an established network, this method focuses on optimising the network to carry out a certain task.

Supervised learning approaches are widely used in machine learning and are known to yield high-quality results. They rely on datasets consisting of the target variable to be predicted. These techniques make use of many variables when

performing classification tasks or continuous values for regression. However, when faced with unfamiliar circumstances, artificial intelligence (AI) systems encounter difficulties. Because of this, the reliability and caliber of the data source have a significant impact on the algorithms' correctness and consistency. Numerous studies comparing the efficacy of AI and human diagnosticians have been conducted as a result of the advancement of AI in medical diagnostics. In order to identify and analyse information for each image, these studies frequently call for the evaluation of hundreds of measurable features, which is more than the human retina can handle. This means that significant processing power is needed. The advancements made in the field of ML has improved the prediction accuracies beyond human. Remarkably, deep learning models have proven to be as good at diagnosing diseases as medical professionals in several applications (Liu X et al. 2019). Examples of these applications include the detection of breast and skin cancer (Lotter W et al. 2021) and skin cancer (Esteva A et al. 2017). For instance, a recent study (Nikolov S et al. 2018) showed the extent to which a deep learning model could split up organs in CT scans of the head and neck with an accuracy that was comparable to that of skilled radiologists. The emergence of many hybrid learning algorithms suggests an increasing convergence of supervised and unsupervised learning. The hybridization of various learning techniques incorporates tactics that tend to offers practical solutions to real-world problems.

This article focus on considering the solutions achieved by employing the conventional and advanced deep learning and machine learning algorithms in addressing issues that requires medical imaging, analysing their respective performances and presenting the top-performing model.

2. BACKGROUND

This section depicts the various application areas, where computer vision can be employed to address issues that require medical imaging.

2.1 Computer Vision

In the field of computer science, computer vision focuses on the identification and understanding of objects on digital images or videos. Computer vision segments the digital image into various portions that represents objects. The segmentation process in computer vision is applied in healthcare domain to identify regions representing diseases, tissues and different anatomical traits. Deep learning models like U-Net, Mask R-CNN and other techniques have been widely employed in medical image segmentation and have shown a significant performance. The digital images used

by these techniques in identifying diseases includes xray images, MRI images, CT scan and histopathological images (Barrag´ an-Montero Ana et al., 2021)

2.3 Medical Imaging

Clinical evaluation and therapeutic analysis in relation to the complexities of human anatomy is achieved through medical imaging. Images like MRIs, CT scan and positron emission tomography (PET) are commonly adopted clinical settings. Clinical decisions about medical or health condition of a patient requires thorough understanding of human anatomy, physiology and the metabolic process. One of the major challenge faced when dealing with medical images for diagnosis is the radiation exposure especially when handling pediatric patients (Armanious et al., 2020). Secondly, the cost of obtaining these images is high (Zhan et al., 2021).

Medical image synthesis looks like a workable answer to these issues since it translates a single visual mode onto another and facilitates conversion across modalities. This method may reduce scanner times, boost the value of pictures acquired, and reduce operating expenses, especially those associated with radiotracers (Wang et al., 2021). Medical image synthesis is thus very beneficial for an extensive variety of health care applications, including MRI-only radiotherapy treatment planning, PET/MRI scanning, picture segmentation, and image enhancement (Armanious et al., 2020).

3. FOCUS OF THE ARTICLE

With the advancement in medical imaging technology, the demand for more sophisticated computer vision techniques has experience a huge surge. Several deep learning models in computer vision have recorded impressive improvements in health care image generation and analysis (Wang et al., 2021). These advancements have performed well on detection of diseases in medical images irrespective of their sources. Most of these techniques use data from existing projects to improve the performance of similar tasks with adequate modifications (Spadea et al., 2019). Therefore, these benefits have speed-up the practical application of health care imagine analysis and increased performance standards in general.

The increased focus on the application of computer vision in medical treatment is attributed to the huge growth and continuous evolution of medical image databases. To begin with, medical images are often rather huge. The digitalization of histopathology slides, for instance, results in gigapixel images, which are roughly $100{,}000 \times 100{,}000$ pixels. In comparison, visual inputs that are around 200×200 pixels are processed by standard convolutional neural networks (CNNs). In addition, different chemical

preparations result in different slides that show the same tissue sample. Furthermore, different digitization tools or configurations can provide different images for the same slide. Radiological modalities, like CT and MRI, generate similarly large 3D images, therefore conventional CNNs have to work with a set of 2D slices or require structural modifications to handle 3D data.

3.1 Cardiology

Deep learning is essential to many healthcare workflows and applications, especially those that involve screening and diagnosis. The most common imaging technique in cardiovascular medicine is cardiac ultrasonography, sometimes known as an echocardiogram.

3.2 Pathology

Within the field of medicine, pathology focuses on the examination and diagnosis of disease. Clinical pathology includes examining surgically eliminated body parts, fluids from the body, tissues (biopsy samples), and occasionally the entire body (autopsy). The macroscopic anatomical composition of a body specimen, the appearance of the cells utilizing chemical and immunological signs, genetic research, and gene markers are some of the aspects that may be taken into account.

3.3 Dermatology

Deep learning (DL) is crucial to a number of significant clinical dermatology initiatives. These duties involve giving specific diagnostic tests for lesions, identifying innocuous from worrying lesions, and assisting in monitoring the evolution of lesions over time. Studies have indicated that convolutional neural networks (CNNs) have the potential to match the performance of board-certified dermatologists in distinguishing either cancerous and benign skin cancers.

3.4 Ophthalmology

Over the past few years, ophthalmology has seen a significant increase in AI-related initiatives, as seen by a number of articles demonstrating clinical diagnostic and analytical capacity that exceed human capabilities. The potential clinical implications are significant; considering the mobility of eye examination equipment, the possibility of setting up mobile clinics and telemedicine might provide testing services to underprivileged areas. In this field, optical coherence tomography (OCT) and fundus

imaging are the main diagnostic tools that are essential for patient diagnosis and treatment.

3.5 Few Challenges in Handling Medical Images

It takes a lot of time and specific knowledge to annotate medical photos for deep learning models. To win over healthcare professionals' acceptance and trust, it is essential to make sure that the model outputs are comprehensible and have practical significance. Medical imaging research and its practical use both heavily depend on managing sensitive patient information and following privacy requirements. considerations.

4. SOLUTIONS AND RECOMMENDATIONS

4.1 Methods for Medical Image Analysis

The last 10 years have witnessed a proliferation of articles showcasing innovative algorithmic capabilities or creative modifications of already existing ones due to the intense study into artificial intelligence (AI) strategies for healthcare applications, especially in ML and DL. Remarkably, convolutional neural networks (CNNs) have gained prominence lately, taking the place of more traditional machine learning methods like random forests (Fig. 2) and support vector machines (SVMs) (Fig. 1).

Furthermore, from 2018 onward, there has been a swift increase in the utilisation of additional DL methods like generative adversarial networks (GANs) and reinforcement learning algorithms. As a result, we explore the foundational methodological aspects of two extensively applied algorithms (random forests and CNNs), alongside the increasing significance of GANs. Furthermore, we present recent instances of these methods applied in medical image processing to underscore their utility.

4.2 Convolutional Neural Network

Computer vision (CV) (Al-Oraiqat et al., 2022), a facet of deep learning, involves the direct identification of objects from raw image data. Initially, experts annotate each image with the accurate diagnosis, establishing the 'ground truth. Following this, computer vision algorithms concentrate on tasks such as recognising images and videos, covering activities such as object classification, detection, and interpretation for predefined categorizations. Convolutional neural networks (CNNs) represent deep learning architectures specialised in processing input images. They prioritise different elements within an image to distinguish one from another. The design

Figure 1. Support Vector Machine

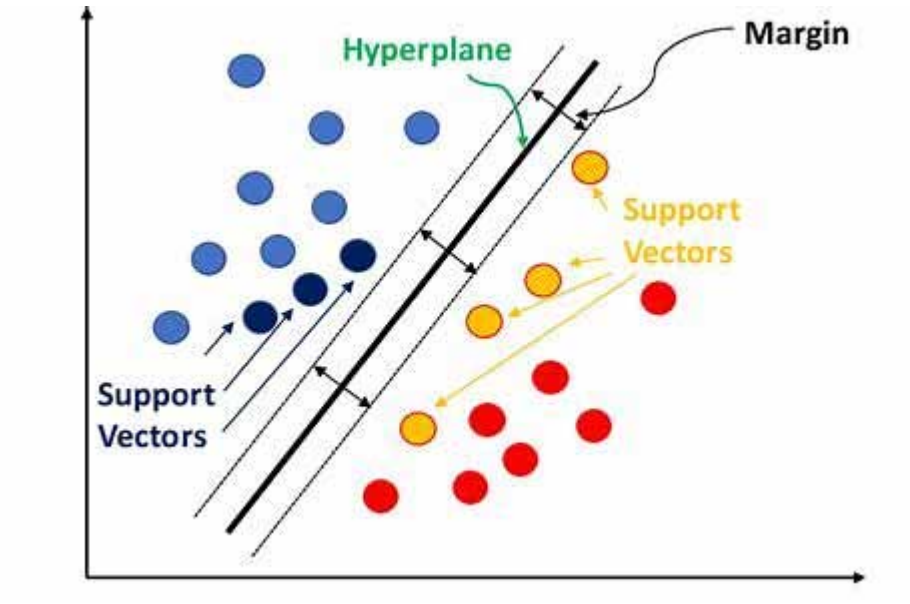
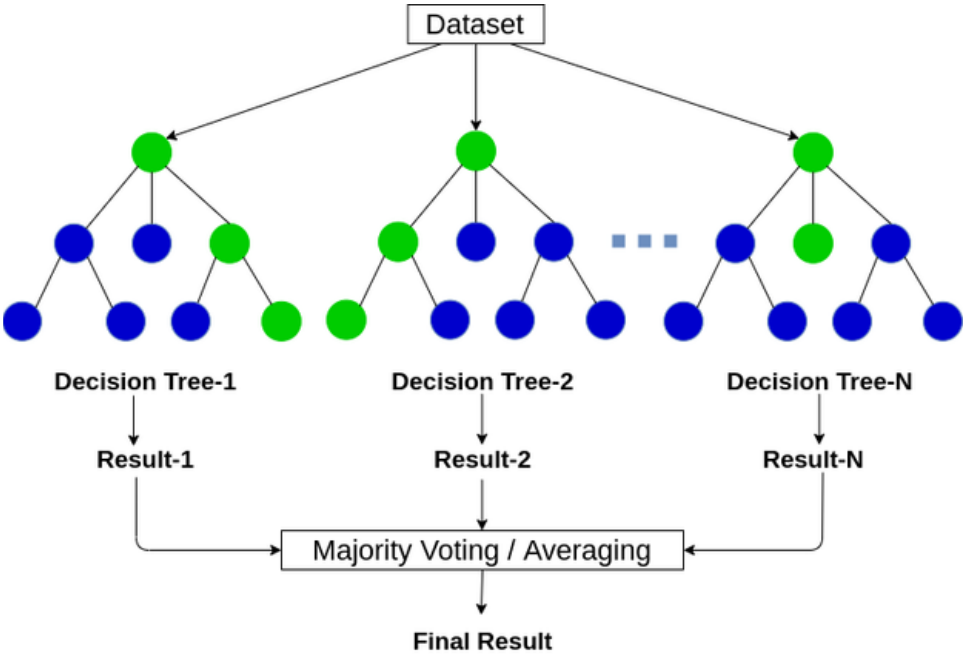


Figure 2. Random Forest

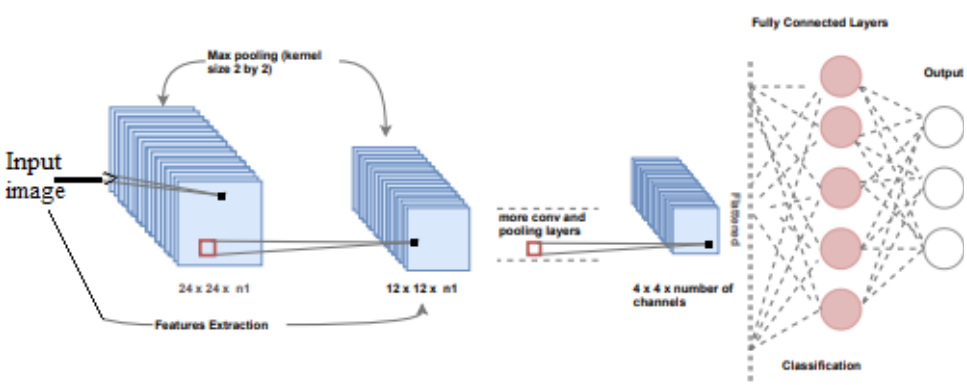


of CNNs reflects the connectivity patterns observed in the neurons of the human brain, operating on a hierarchical model that forms a funnel-shaped framework. This arrangement culminates in a fully connected layer where all neurons interconnect and process output. Notably, within a historical context, the accuracy achieved by state-of-the-art computer vision has been favourably compared to human accuracy in tasks like object classification.

CNN-based methods have significantly accelerated advances in computer vision (CV), particularly in areas such as medical analysis of images and categorization. Although they have been used since the 1980s, the current wave of computer power and algorithmic improvements has made CNNs indispensable for a wide range of vision-related applications. By utilizing weight sharing and partially connected layers, CNNs are able to capture the inherent representation of pictures. As seen in Figure 3, many CNN architectures usually consist of a series of convolution layers, followed by activation functions and pooling layers. These networks produce activation maps, also called feature maps, by repeatedly applying filters to input images. These maps identify important regions within the input image. CNNs and other deep learning (DL) approaches are largely responsible for improvements in object detection and recognition inside images. These developments have proven essential, especially in fields like robotic surgery and medical applications (Esteva A et al. 2019). In these situations, the difficulty goes beyond simple image classification and involves abnormality localization within medical imaging. Not only must an image be classified as abnormality-free, but the position of these anomalies within the medical image must also be accurately identified and defined.

Radiology is one of the many fields that have shown interest in the Convolutional Neural Network (CNN) (Such, F.P. et al. 2019), a significant class of artificial neural networks that is extensively used in computer vision. CNNs are designed to use

Figure 3. CNN model architecture



backpropagation to independently and adaptably understand the spatial hierarchies of features. As illustrated in Fig. 3, they use a variety of basic elements, including convolution layers, pooling layers, and fully linked layers, to accomplish this adaptive learning process. CNNs have the ability to learn using both supervised and unsupervised techniques. They are useful for medical image analysis because they are specifically designed to take use of spatially structured information. In such images, adjacent pixels representing a particular anatomical structure often share similar intensity characteristics. Moreover, these networks utilise specific parameterized neuron patches to identify pixels that correspond to specific anatomical regions.

4.3 Convolution

Convolution stands for a specialised linear operation critical for extracting features. It entails applying a kernel, a small numerical array, onto an input tensor, which is essentially an array of numbers. At each position of the tensor, there's an element-wise multiplication between the kernel and the input tensor, followed by a summation of these products, resulting in an output value positioned in the corresponding location of the output tensor. This resulting tensor is termed a feature map. This process is repeated by employing multiple kernels, generating an array of feature maps that portray distinct characteristics of the input tensors. Each kernel serves as a unique feature extractor. The convolution operation is determined by two crucial hyperparameters: the size and quantity of kernels. Typically, the size ranges from 3×3, occasionally extending to 5×5 or 7×7. Meanwhile, the number of kernels is flexible and influences the depth of the output feature maps.

4.4 Pooling Layer

A pooling layer aids in a standardised down sampling process that reduces the dimensionality of feature maps within the plane. This reduction intends to introduce translation invariance to minor shifts or distortions while simultaneously decreasing the number of subsequent learnable parameters. It's important to note that pooling layers lack any learnable parameters. Instead, parameters like filter size, stride, and padding serve as hyperparameters in pooling operations, mirroring the arrangement found in convolution operations.

4.5 Max Pooling

One of the frequently adopted pooling operations is max pooling, which involves selecting patches from input feature maps and retaining only the highest value within each patch, discarding other values. In application, a frequently used method

involves employing a 2×2 filter size alongside a stride of 2, effectively reducing the in-plane dimension of feature maps by half. It's important to note that while the height and width dimensions undergo reduction, the depth dimension of the feature maps remains unchanged.

4.6 Fully Connected Layer

The resulting feature maps from the last convolution or pooling layer are typically flattened, converting them into one-dimensional arrays or vectors of numerical values. These vectors are then connected to one or more fully connected layers, often termed dense layers. Within these dense layers, every input is linked to every output through adjustable weights. Following the extraction of features and down sampling by convolution and pooling layers, a subset of fully connected layers translates these features into the network's final outputs, such as class probabilities in classification tasks. Usually, the final fully connected layer comprises the same number of output nodes as there are classes in the given problem. Each fully connected layer is combined with a non-linear activation function, like ReLU, to introduce non-linearity into the network.

4.7Activation Function

The activation function applied in the final fully connected layer is typically distinct from the preceding layers. Choosing the right activation function is pivotal and relies on the task's specifics. In the context of a multiclass classification task, one commonly used activation function is the softmax function. This function normalises the real-valued outputs from the last fully connected layer into probabilities for target classes. These probabilities span from 0 to 1, summing up to 1 collectively, effectively representing the likelihood of each class. The CNN model used in this paper consists of consists of five convolution and pooling blocks. Every block has a pooling layer after two convolution layers. Table 1 displays the configuration of the CNN.

4.8 Training and Loss Function

Training is an operation during which the neural network adjusts its hyperparameters, the weights, and biases, and the model maps the link between the training input and the outputs. The neural network loads each training input through a procedure known as forward propagation. Backpropagation is the process of comparing the output that the model predicts to the target output that has been provided. A loss function

Table 1. Structure of the CNN Model

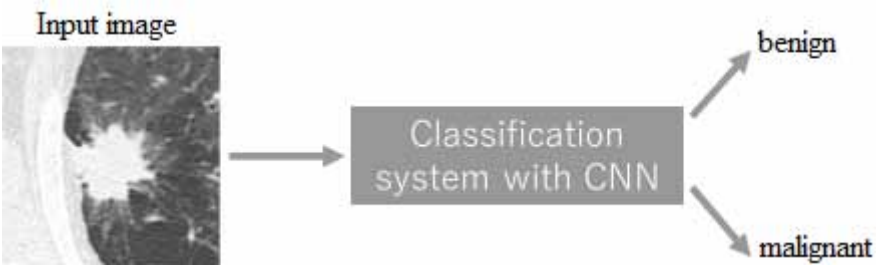
Layer	Type of Layer	Kernel size
Convolution Layer	Conv2D	3 x 3
Convolution Layer	Conv2D	3 x 3
Pooling Layer	MaxPool	2 x 2
Convolution Layer	Conv2D	3 x 3
Convolution Layer	Conv2D	3 x 3
Pooling Layer	MaxPool	2 x 2
Convolution Layer	Conv2D	3 x 3
Convolution Layer	Conv2D	3 x 3
Pooling Layer	MaxPool	2 x 2
Convolution Layer	Conv2D	3 x 3
Convolution Layer	Conv2D	3 x 3
Pooling Layer	MaxPool	2 x 2
Convolution Layer	Conv2D	3 x 3
Convolution Layer	Conv2D	3 x 3
Pooling Layer	MaxPool	2 x 2
FC Layer	-	-
Dense Layer	-	-

evaluates how well a neural network models the training set of data by comparing the target and predicted output values.

4.9 Classification

In the field of medical imaging, deep learning algorithms are widely applied to classify identifiable lesions of interest in healthcare images into distinct categories. As Fig. 4 illustrates, lung nodules in computed tomography (CT) images are commonly identified using deep learning approaches, which help distinguish between benign and malignant features. For convolutional neural networks (CNNs) to perform successful categorization, a big dataset containing tags matched to training data is essential. The first training set for the lung nodule classification case consists of labels for lung nodules found in CT scans.

Figure 4. Classification with CNN



4.10 MRI Heart Analysis

The heart is a vital organ that has a significant impact on human health. It acts as the body's perpetual pump, circulating blood throughout it to support vital organs and maintain steady levels of energy. A short cardiac arrest could have potentially lethal consequences. Thus, for those with cardiovascular diseases (CVD), efficient cardiac function monitoring is essential.

In recent decades, various medical imaging modalities have emerged as powerful tools for visualising and assessing the heart's health. These include:

1. **Echocardiography (Ultrasound):** This technique uses ultrasonic waves to provide real-time images that show the heart's anatomy and operation. This portable, non-invasive method is ideal for bedside monitoring and the early detection of cardiovascular disease (CVD). Figure 5 illustrates it.

Figure 5. Echocardiography image of the heart

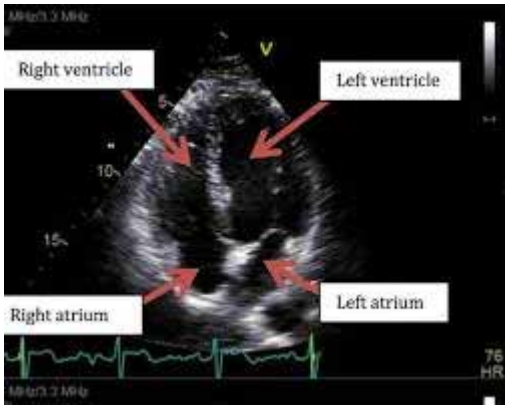
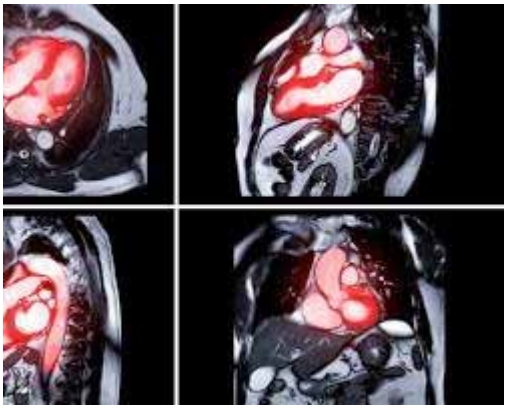


Figure 6. CT scan of the heart

2. **X-Ray Computed Tomography (CT):** This imaging method creates fine-grained cross-sectional pictures of the heart and its surrounding components using X-rays. It offers important details regarding the structure of the heart, such as if coronary artery blockages exist. It is seen in Figure 6.

3. **Magnetic Resonance Imaging (MRI):** This modality uses radio waves and magnetic fields to provide incredibly detailed images of the blood flow and soft tissues of the heart. As seen in Fig. 7, it provides exceptional contrast resolution and permits quantitative investigation of heart activity.

Figure 7. MRI scan of the heart



Every imaging modality has benefits and drawbacks of its own, and the particular clinical question and patient requirements frequently dictate which technique is best. For instance, because echocardiography is accessible and reasonably priced, it is frequently utilized for a preliminary evaluation of heart function. MRI, however, can be favored in complex instances for a more thorough assessment of cardiac tissue and blood flow.

The field of cardiovascular medicine has changed as a result of the development of advanced imaging technologies as well as advancements in AI and machine learning. These developments have led to:

- Improved diagnosis: More accurate and early detection of cardiovascular diseases, leading to better patient outcomes.
- Personalized medicine: Customized treatment strategies designed considering individual patient attributes and the severity of the disease.
- Minimally invasive procedures: Development of new and less invasive procedures for treating heart conditions.

Medical imaging is still essential for the diagnosis, treatment, and management of cardiovascular illnesses because it offers important insights into the structure and function of the heart. This helps to enhance patient care and quality of life.

A crucial technique in contemporary medicine, magnetic resonance imaging (MRI) provides a secure, non-invasive means of viewing the internal organs of the human body. In contrast to CT scans, which use radiation, magnetic resonance imaging (MRI) uses strong magnets and the spin of hydrogen atoms to produce detailed images. As the second-safest imaging modality available, MRI is particularly safe for patients with radiation sensitivity or frequent imaging requirements.

4.11 Segmentation of Organs in Medical Images

Accurately segmenting anatomical features and tissues is crucial for computer-aided diagnosis tasks in medical image analysis as well as quantitative parameter assessment. In the past, convolutional neural networks (CNNs) have relied on a patch-based approach to segment images. The model is trained to predict the class label of the central pixel in each patch after an image has been broken into smaller patches. This approach has benefits, but it also has disadvantages.

Redundancy and Inefficiency: One significant drawback of the patch-based approach is its inherent repetition in visual coverage. The model has to analyse the same area of the image multiple times since every patch intersects with its neighbours. This duplication leads to inefficient use of training and resources.

Trade-off between Localization and Context: Patch size affects the trade-off between contextual information and localization accuracy. Larger patches require larger pooling layers, which preserve context by spanning a larger area but may decrease localization accuracy. Conversely, tiny regions lack context but offer greater localization because of their narrower scope. This fundamental trade-off needs to be carefully taken into account when configuring CNNs for medical image segmentation, particularly for tasks like cardiac image segmentation where accurate object localization is crucial.

Pixel-Wise Segmentation: Recent Advances: While traditional patch-based CNNs perform well for object localization, they are not the best for pixel-wise segmentation. To get around this restriction, recent work has focused on modifying traditional CNNs to support complete pixel-wise segmentation. The ultimate aim of these modifications is to generate segmentation outcomes that are more dependable and precise by striving to integrate both local and global context.

Examples of Recent Advances:

- U-Net: This design uses an encoder-decoder framework, where the last segmentation filter is generated by the decoder progressively up sampling the features that the encoder captures at different scales.
- Atrous convolutional layers are used in this system to preserve high resolution while capturing long-range context.
- Mask R-CNN: This model provides bounding boxes and pixel-level masks for segmented objects by combining object recognition and segmentation.

These developments demonstrate how CNNs for medical picture segmentation are constantly evolving and present viable ways to get over the drawbacks of conventional patch-based methods and produce precise, pixel-by-pixel segmentation results.

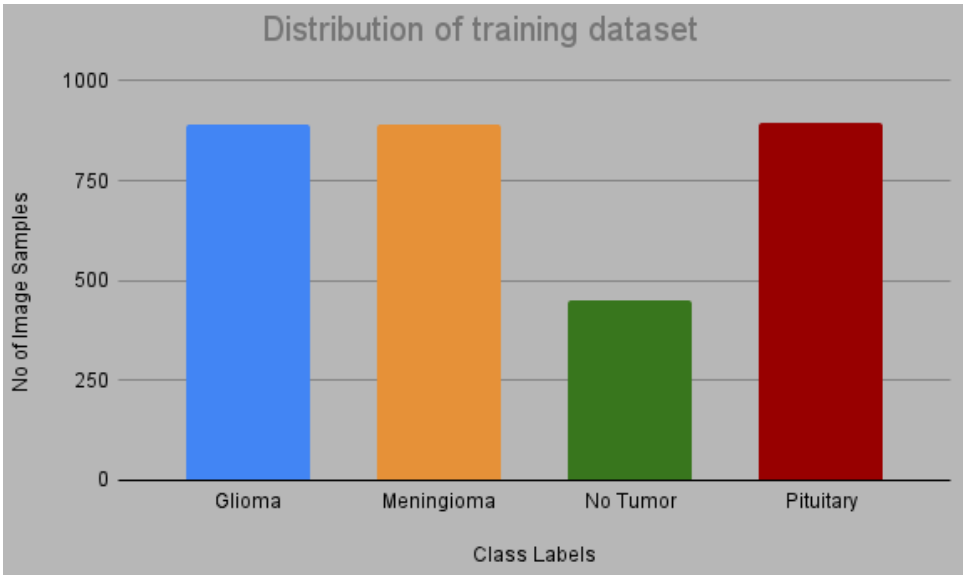
5. COMPARISONS OF DEEP LEARNING MODELS IN MEDICAL IMAGE CLASSIFICATION

This section compares the performances of different deep learning techniques applied to MRI image datasets of brain tumors. The techniques were applied to the tree and other datasets for both segmentation and classification.

5.1 Medical Images Data Collection

The Kaggle repository has made the brain tumor MRI dataset available as an open-source resource. The information is comprised of an assortment of various human

Figure 8. Distribution of Dataset for Model Training



brain MRI scans from Figshare, the SARTaj dataset, and Br35H. The dataset consists of 7023 mages of human brain MRI images belonging to four classes: glioma, pituitary, non-tumor, and miningioma. The no-tumour class images were collected from Br35H, and the other classes were collected from the figshare and SARTaj datasets for model training.

5.2 Evaluation Metrics

The performances of the models are determined by the outcomes derived from the prediction results from the individual models. The criteria employed include accuracy, precision, recall, and F1. Although accuracy is deemed significant, it should be more than solely relied upon to assess performance. Therefore, the evaluation also considers recall, precision, and F1-score, contributing significantly to performance evaluation. These metrics are highly beneficial for gauging performance effectively.

- Accuracy: to evaluate the performance of the model.

  Accuracy = (TP+TN)/(TP+FP+TN+FN)

- Recall: To evaluate the ratio of correctly diagnosed versus total.

Recall = TP / (TP+TN)

- Precision: To evaluate how many cancer images are diagnosed ideally.

Precision = TP / (TP+FP)

- F1-Score: This value is the harmonic mean of precision and recall and is considered a better performance indicator.

F1 Score = (2xPrecision x Recall) / (Precision + Recall)

The above formulas used for calculating the models performance presented in Table 2. The term T.P., TN, F.P., and F.N. refers to a true positive, true negative, false positive and false negative

5.3 Performance Comparison

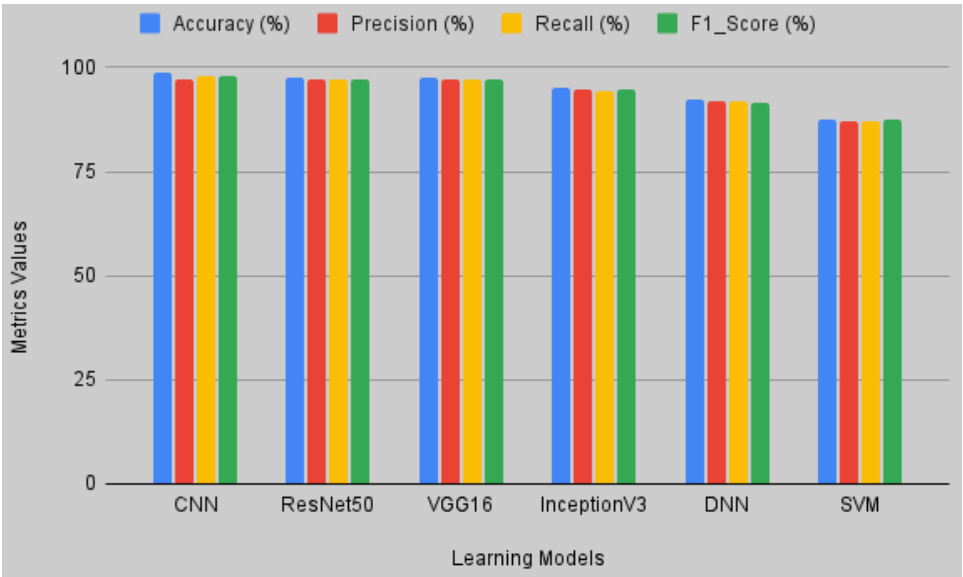
In this section, the performance of the models based on accuracy, precision, recall and f1_score is presented. Based on the results, Convolutional Neural Network outperforms the other conventional Machine Learning and Deep Learning approaches.

Figure 9 shows the performance of the classification models based on prediction accuracy, precision, recall and f1_score measures. Based on the figure, it can be seen that the conventional CNN model outperforms the ResNet50, VGG16, InceptionV3, DNN and the conventional Support Vector Machine (SVM).

Table 2. Performance of the classification models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1_Score (%)
CNN	98.7	97.2	98	98.1
ResNet50	97.4	97.1	97.0	97.0
VGG16	97.4	97.0	97.2	97.1
InceptionV3	95.1	94.7	94.5	94.7
DNN	92.4	91.9	92.0	91.5
SVM	87.5	87.2	87.2	87.4

Figure 9. Performances of the Classification Models



6. CONCLUSION

In this work, we present a comprehensive overview of the principal neural network architectures, advanced building-blocks, and loss functions employed in the realm of cardiac image segmentation. The past decade has witnessed a notable surge in machine learning applications within cardiac research. Initiatives like euCanSHare (http://www.eucanshare.eu/) have played a pivotal role in establishing international, multi-cohort platforms for cardiovascular studies. These initiatives have been instrumental in fostering innovation and enhancing the potential for clinical implementation. Within this landscape, cardiac image segmentation stands as a crucial component, striving to achieve a personalized, patient-centric, precise, and time-effective approach in diagnosing and managing cardiac pathologies. Notably, image segmentation in the biomedical machine learning domain has garnered substantial clinical acceptance by alleviating clinician workload while minimally impacting their decision-making process. In future, the CNN model will be trained and generalized on various medical image datasets to enable its use in different medical sub domain.

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